

OPUS.NET - Visible Conversation Transcript

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USER

MEL and ENTER needs a .NET multi-tier framework. let us start the process of designing this. i call it OPUS.NET. it will have a object-network database persistence layer, a middle-tier layer where business rules close to the physical server machine will be handled including queries and multi-machine distributed storage and distributed transaction management will be handled, a server-side facade into the middle-tier domain model, where this facade packs all requests in a binary TCP/IP protocol for the client-side of the domain model to use and finally a way to choose how much of the server side domain model content should be downloaded into the client side domain model. engage

ASSISTANT

Yes. The architecture is coherent, but the key is to define OPUS.NET as a distributed domain-object runtime, not merely a networking or ORM framework. I would start by separating the responsibilities sharply enough that MEL and ENTER can share the same infrastructure without either product leaking into the framework.

Yes. I think OPUS.NET should be designed as a reusable distributed object-network runtime for .NET, with MEL and ENTER simply becoming domain applications built on top of it.

The most important architectural decision is that OPUS.NET should not expose database concepts to the application. Applications should operate on live domain objects and relationships. Persistence, distribution, synchronization, serialization, caching, transactions, and client/server transport happen underneath that domain model.

OPUS.NET - Initial Architecture

I would divide it into six major layers:

APPLICATION DOMAIN

MEL / ENTER / OPUS ERP / etc.

6. CLIENT DOMAIN RUNTIME

Local object network / cache / lazy loading / change set

5. OPUS.NET BINARY PROTOCOL + SERVER FACADE

TCP/IP / authentication / requests / responses / streams

4. SERVER DOMAIN RUNTIME

Domain objects / business rules / object lifecycle

3. DISTRIBUTED MIDDLE TIER

Queries / transactions / routing / shards / coordination

2. OBJECT NETWORK ENGINE

Object identity / relations / serialization / indexing

1. PHYSICAL STORAGE

GUID -> binary object blob / indexes / transaction journal

This is slightly different from the conventional UI -> API -> service -> ORM -> SQL architecture. That is intentional.

OPUS.NET's fundamental abstraction becomes:

A persistent distributed network of addressable .NET objects.

1. Object-network persistence

Your original concept of an object fundamentally being:

GUID

SerializedBlob

is very attractive as the canonical storage representation.

For example:

8FAD2B6A-... -> [serialized Customer object]
91BDE388-... -> [serialized Order object]
CB11994E-... -> [serialized Product object]

And relations are simply GUID references.

OPUS.NET reconstructs an ordinary .NET object network in memory.

That gives you something quite different from a relational database.

The programmer thinks:

order.Customer.Name

rather than:

SELECT ...
FROM Orders
JOIN Customers ...

But we need one important addition

A pure GUID + Blob database is excellent for loading objects by identity, but terrible for queries.

We should therefore keep the canonical object storage simple:

ObjectStore

ObjectId GUID
Blob BINARY

while OPUS.NET maintains separate derived structures:

TypeIndex
PropertyIndex
RelationIndex
FullTextIndex
TransactionJournal
ShardDirectory

These structures can always theoretically be reconstructed from the blobs.

The object blob is truth. Indexes are accelerators.

[The remainder of this first assistant response continues through the architecture discussion: Object Network Engine, distributed middle tier, transactions, server-side domain model, server facade, Domain Projection, loading modes, server-side queries, client-side domain runtime, change sets, versioning, stream separation, five fundamental concepts, proposed solution structure, and the concluding architecture diagram.]

USER

i understand the basic architecture. let us begin with the server side persistence framework, the DB itself. i see the simplest solution being a object network db where each object (identified with a guid and the object content serialized as a blob) is a file on the file system where the filename is the guid and the postfix ".bin" all stored in a single folder. the middle tier that knows the guid of the object (stored in memory in the server-side domain model) simple accesses each file and CRUD this file. analyze

ASSISTANT

Yes. As a first persistence engine, this is extremely attractive because it reduces the database to the smallest possible abstraction:

Object identity = GUID
Persistent representation = serialized binary blob
Physical address = <root>\<GUID>.bin

So if the in-memory domain model holds a GUID, the persistence layer can map it directly to Objects\<GUID>.bin and CRUD becomes almost trivial.

The core contract could be as small as:

```
public interface IObjectStore
{
    bool Exists(Guid id);
    byte[] Read(Guid id);
    void Create(Guid id, byte[] data);
    void Update(Guid id, byte[] data);
    void Delete(Guid id);
}
```

Above that, serialization turns it into a typed object store:

```
T Load<T>(Guid id);
void Save<T>(Guid id, T obj);
```

Its biggest strength is that object identity and physical persistence match almost perfectly.

There is no ORM, no impedance mismatch, no table decomposition, no joins, no foreign keys and initially no database server process.

The persistence layer does not need to understand the meaning of relationships. It simply knows GUID -> file.

CRUD becomes beautifully simple: serialize/write, read/deserialize, serialize/replace, delete.

The main issues identified were: single-folder scalability, atomic file replacement, optional metadata, serialization strategy, GUID references rather than recursive graph serialization, OS file caching, query limitations, concurrency, deletion/recovery, and separating serializer/store/physical store.

The fundamental idea remains sound: Every persistent object has exactly one independently addressable object file, identified by GUID; the file store is free to partition those files physically.

USER

source code changes that requires and update of the schema on file will be a manual process to transform v1 domain model schema to v2

```
■ T Load<T>(Guid id);
■ if IObjectStore.Exists(id) then
■   object T = Deserialize(IObjectStore.Read(id))
■ else
■   throw ObjectDoesNotExistException
■ endif
■ void Save<T>(Guid id, T obj);
■ if IObjectStore.Exists(id) then
■   IObjectStore.Update(id, Serialize(T, obj))
■ else
■   IObjectStore.Create(id, Serialize(T, obj))
■ endif
■ void Delete(Guid id);
■ if IObjectStore.Exists(id) then
■   IObjectStore.Delete(id)
■ else
■   throw ObjectDoesNotExistException
■ endif

class ServerDomainModelStorageWithBinarySerialization
■ T Load<T>(Guid id);
■ if IObjectStore.Exists(id) then
■   object T = DeserializeWithSchema(schemaObject, IObjectStore.Read(id))
■ else
■   throw ObjectDoesNotExistException
■ endif
■ void Save<T>(Guid id, T obj);
■ if IObjectStore.Exists(id) then
■   IObjectStore.Update(schemaObject, id, SerializeWithSchema(schemaObject, T, obj))
■ else
■   IObjectStore.Create(schemaObject, id, SerializeWithSchema(schemaObject, T, obj))
■ endif
```

```

    ■■endif
    ■void Delete(Guid id);
    ■■if IObjectStore.Exists(id) then
    ■■■IObjectStore.Delete(id)
    ■■else
    ■■■throw ObjectDoesNotExistException
    ■■endif

```

ASSISTANT

This is a good simplification. Given your stated execution model-one server-domain-model request at a time on a singleton thread-I would deliberately keep this layer very small and avoid adding locking, MVCC, transaction coordination, or schema migration machinery here.

Your two classes are conceptually correct, but I would change the structure slightly so that storage semantics and serialization strategy are separated. Right now the two classes duplicate almost all logic.

The common behavior is Load/Save/Delete; only Serialize/Deserialize changes.

So I would make the architecture:

```

ServerDomainModelStorage
|
+---- IObjectStore
|
+---- IObjectSerializer

```

IObjectStore knows GUIDs and bytes.

IObjectSerializer knows objects and bytes.

ServerDomainModelStorage knows how to combine them.

Domain Model knows objects.

The schema belongs entirely to the serializer, not IObjectStore.

Your schema strategy also makes sense under the development process you described. An explicit conversion can load using V1 schema, transform V1 object -> V2 object, then serialize using V2 schema.

Your single-thread server model also means the usual Exists/Create race cannot occur inside the server domain model, provided only the OPUS.NET server process modifies the object-store directory.

The resulting lowest stack is:

```

SERVER DOMAIN MODEL
|
ServerDomainModelStorage
|
+-- IObjectSerializer
|   +-- NET Serializer
|   +-- OPUS Binary Serializer -> DomainSchema
|
+-- IObjectStore
    +-- FileObjectStore -> GUID.bin files

```

Responsibilities:

Domain Model = behavior and relationships

ServerDomainModelStorage = persistence semantics

Serializer = object <-> bytes

ObjectStore = GUID <-> bytes

FileObjectStore = GUID <-> physical .bin file

USER

So, i could implement SQLServerObjectStore implementing IObjectStore for implicit transactions and crash recovery?

ASSISTANT

Exactly. That is one of the main benefits of keeping IObjectStore this narrow.

You can have FileSystemObjectStore and SQLServerObjectStore implementing the same interface.

For SQL Server, the physical schema could be almost absurdly simple:

```
CREATE TABLE Objects
(
    ObjectId UNIQUEIDENTIFIER NOT NULL PRIMARY KEY,
    Data VARBINARY(MAX) NOT NULL
);
```

Then:

Exists(Guid) -> SELECT existence

Read(Guid) -> SELECT Data

Create(Guid, bytes) -> INSERT

Update(Guid, bytes) -> UPDATE

Delete(Guid) -> DELETE

The major advantage is precisely what you identified: SQL Server gives the lowest storage layer mature transactional durability and crash recovery without changing the OPUS.NET object model at all.

SQL Server gives an implicit transaction around each individual statement. If a domain request performs several object operations, those are separate SQL operations unless wrapped in an explicit transaction.

So there are two levels: storage-operation atomicity and domain-operation atomicity.

OPUS.NET is not becoming a relational persistence framework. SQL Server is merely being used as a highly reliable GUID -> binary-blob storage engine. OPUS.NET retains ownership of the object model, relationships, serialization and later querying semantics.

USER

let us analyse further under these conditions: the server side domain model is completely loaded in memory (except from the additional BLOB storage mechanism that will be available on the side of the domain model) and based on f.ex domain model types distributed across multiple server machines, hard-coded or configured in a server-side configuration file, etc something, like even a frontend server-side load balancing mapper to a secondary layer of multiple domain model server machines....bla.bla. this and the following: how do we handle queries? first thing that pops up is that the facade of the server-side domain model offers explicit Find-methods as a flat list on a singleton object, but also more advanced such as server-side domain model uses an in-memory index to speed things up. analyse

ASSISTANT

Yes. Under those conditions, queries become much simpler than in a conventional database architecture because the server-side domain model itself is the primary queryable state. Persistence is mainly for restart/recovery, not for normal read access.

Persistent object store = restart/recovery

In-memory domain model = live authoritative state

Query subsystem = navigation/indexing over live state

Normal queries should not hit the persistence layer at all.

The simplest first version is exactly what you suggest: explicit query methods exposed by the server-side facade, such as FindCustomer, FindCustomerByEmail, FindCustomersByCountry, FindOrdersByCustomer, and FindProducts.

The facade becomes the query contract between client and server.

The implementation can initially scan collections, later use an in-memory index, and later route to another server without changing the client-facing method.

I suggested separating domain queries from query mechanics, using typed in-memory indexes where useful, rebuilding indexes at startup, and treating distribution as query routing rather than changing query semantics.

I also suggested three conceptual levels:

1. Domain-specific Find methods.
2. Indexed Find methods.
3. Distributed Find methods.

Large BLOBs remain outside the loaded domain graph and are demand-loaded/streamed.

The key query principle is: Queries operate on the live in-memory domain model. Persistence is not queried. Domain-specific facade methods define query semantics; indexes are transparent accelerators; distribution is transparent routing.

USER

i agree that the registry is necessary for reconstructing the object network after startup, but would not the registry provide a typed object? how do i know what type the object has if retrieved from the registry? my approach as a hierachial domain model with a known root (such as a singleton "Application" object and underneath that the basic practical typed objects, such as a Customer hashlist. am i wrong here? would not storing objects directly in the index/registry/hashtable/collection be smaller and faster then the indirect mapping? you seem excited and are getting slightly ahead of the design process here.

ASSISTANT

You are not wrong. Given the architecture you are describing, your hierarchical root-object approach is probably simpler than the generic global Guid -> object registry I introduced.

If the server domain model is fully loaded and structurally known, you can have:

```
Application
{
    Dictionary<Guid, Customer> Customers;
    Dictionary<Guid, Order> Orders;
    Dictionary<Guid, Product> Products;
}
```

Then the collection itself is both registry and typed index.

```
Customer c = Application.Customers[id];
```

already answers both questions: what object has this GUID, and what type is it.

The generic registry was implicitly solving the case where only a GUID is known and nothing else. Your current domain model does not necessarily have that requirement.

Startup can load the Application singleton, load Customers/Orders/Products into typed collections, and resolve relationships using the known typed property.

Storing actual object references directly in in-memory indexes is preferable when those indexes exist only in memory.

So the rule becomes: Persist relationships as GUIDs; once loaded, resolve them to normal typed object references. In-memory indexes may point directly to the typed objects.

I therefore retracted the claim that a universal ObjectRegistry<Guid, object> is necessary at this stage. Your root-domain hierarchy already provides the registry function in a more strongly typed and domain-natural way.

USER

i can not see value in a non-typed registry since in the end, the client ui must map data to a user-exposed structure, even free-form ui interface needs this. f.ex the binary content of an untyped object, will this be understood by the end user. the only use i see is some kind of diagnostics where the untyped is examined though reflection, but what will that do to practical enterprise business use?

ASSISTANT

I agree with your conclusion. For the architecture you are defining, a global untyped registry has little practical value and would mostly introduce indirection without solving a real business problem.

Somewhere in the chain, the data must acquire domain meaning. A user does not interact with an arbitrary serialized blob. They interact with a Customer, Invoice, Campaign, Product, Payment, etc.

Dictionary<Guid, object> does not really remove typing. It merely postpones it.

A stronger principle is appropriate:

Every persistent business object belongs to a known domain type, and every in-memory object is reached through a typed part of the domain model.

Reflection can still be useful for diagnostics, serialization, UI generation, schema inspection, logging, or OPUS Designer integration without requiring an untyped registry.

A dynamic UI is not an untyped domain. A dynamic UI can be generated from an extremely strongly typed domain model.

So the global untyped registry can be dropped at this stage. Typed collections under Application can provide ownership/containment, primary GUID lookup, iteration, startup reconstruction target, and domain navigation, with additional typed indexes where query performance requires them.

USER

my idea is that the server-side domain model is NOT exposed directly to the client, but every FUNCTION needed by the client (FindByEmail, SaveCustomer, etc) is offered as a facade, i.e a flat list of functions. the implementation of these function will manage transactions, load balancing, reader/writer lock, etc

ASSISTANT

Yes. That is a clearer boundary, and I think it is stronger for your architecture.

The client should not navigate or manipulate the server-side domain model directly at all. The server-side domain model is an implementation detail. The client sees only a facade contract consisting of explicit functions.

The key rule is:

The facade defines what operations are legal; the domain model defines how those operations work.

For example, SaveCustomer can internally route the request, acquire write access, begin a transaction, validate/update the Customer, update indexes, persist changes, commit, release write access, and return success.

FindCustomerByEmail can route, acquire read access, query an in-memory email index, map the result, release access, and return.

The client knows none of that.

The flat list is the service contract. It does not imply one giant facade class; internal implementation can be distributed across classes, assemblies, or machines while the remote surface remains flat.

The server-side domain model is private. Clients never access it directly. All remote interaction occurs through explicit facade functions.

USER

what is DTD? the reason for defining a flat list of functions is that this has to be mapped to a binary TVP/IP protocol interfacing the client and the server and flat is the simplest rather than an unneeded hierarchy of functions that would be the result of namespaces. the namespace organization will be implicit in the function naming, f.ex "FindOrderOwnedCustomer". i did not specify that the functions would be owned by one facade class, but a single tier available to the tcp/ip server host entry point (as seen by the client). you are still a bit ahead of the design process.

ASSISTANT

DTD usually means Document Type Definition, historically used with XML to define what elements and structure an XML document may contain. It is not relevant to the OPUS.NET design you are describing unless XML enters the picture later.

Your clarification changes how the facade should be framed. You are not proposing one giant Facade class or a hierarchical service namespace. You are defining a single callable server tier, exposed through the TCP/IP host entry point, where every callable operation has a unique flat function identity.

From the client's point of view, the contract is simply names such as:

FindCustomerByEmail
FindOrderOwnedCustomer
SaveCustomer
DeleteOrder
CreateCampaign

The binary protocol can map:
FunctionId
RequestPayload

to a flat dispatch table:
FunctionId -> implementation

The implementation may physically live across many classes, assemblies, machines, or domain partitions. That organization is invisible to the client and protocol.

Flat protocol surface does not imply flat source-code organization.

The server exposes one flat set of uniquely named callable functions. The TCP/IP protocol maps requests directly to those functions. How those functions are implemented or grouped internally is not part of the client contract.

USER

we now have the persistence layer, the server-side domain model layer and the function list facade layer, this might seem like a lot of layers, but imagine the testability of this design compared to merging functionality in fewer layers, which i believe is more common in competing frameworks today. comment on that and let us continue to the tcp/ip layer server-side

ASSISTANT

I agree with the layering here. It is more layers than many mainstream frameworks use, but each layer has a very narrow responsibility, and that gives you unusually clean test boundaries.

Persistence can be tested with no business model at all.

The domain model can be tested entirely in memory.

Each facade function can be tested as a pure server-side operation without opening a socket.

The TCP/IP layer can be tested separately as binary request -> decode -> dispatch correct facade function -> encode response.

A networking bug therefore does not require debugging business logic, and a business-rule bug does not require packet-level debugging.

Layers become harmful when they merely forward calls without adding a meaningful contract. Here the contracts are distinct enough to justify themselves.

The server-side stack is:

```
TCP/IP Server Host
  v
Function-list facade
  v
Server-side domain model
  v
Persistence
```

The TCP/IP server initially needs only to accept a connection, read a complete request, decode function identifier and parameters, dispatch a facade function, encode the result, and write the response.

Given the flat facade, the protocol can be extremely simple around FunctionId and Payload.

The TCP host does not need to understand Customers or Orders. It only needs to map FunctionId to a registered function handler.

The next design question was whether one TCP connection handles one outstanding request at a time or multiple outstanding requests. Given the current server-domain-model model, the simpler request -> worker -> facade -> response flow is sufficient.

USER

again, versioning belongs to the developer using a practical software configuration management methodology with supporting tools to convert from an older version to the recent version. now for the tcp/ip server interfacing directly on the network. there will be one listener thread. the listener reads the first byte of the incoming request package and if this byte maps the boolean value of true or false, it acquires a write-lock, if it is false, it acquires a read-lock. it then hands the incoming package to a worker thread on from a thread pool. the worker thread then reads the next section of the incoming package and identifies which facade function should be called, with the now deserialized input data as function parameters, it calls the facade function, wait for the function call to return and serializes the result into a response and sends it back to the client (knowing the client-side end-point) and goes back to the thread pool.

analyse

ASSISTANT

Yes. This is a coherent next layer, but I would make two important corrections while preserving your basic design.

The TCP layer establishes the concurrency boundary before a facade function is allowed to touch the domain model.

I would move lock acquisition to the worker. If the listener itself blocks on `EnterWriteLock()`, it cannot service incoming connections/requests. The listener should only read/classify/dispatch; the worker owns the lock.

A `ReaderWriterLock` means the execution rule becomes: concurrent read-only facade operations are permitted, while write operations execute exclusively.

The first byte should not be trusted as the sole authority for whether an operation is read or write. The server can maintain metadata for each facade function and verify the requested access mode, or derive access mode from `FunctionId`.

The worker lifecycle:

1. Receive complete framed request.
2. Identify facade function.
3. Determine/verify READ or WRITE.
4. Deserialize parameters.
5. Acquire appropriate domain lock.
6. Invoke facade function.
7. Wait synchronously for return.
8. Release domain lock.
9. Serialize returned value.
10. Send response.
11. Return worker to pool.

TCP itself needs request framing because it is a byte stream. A package length can precede `AccessMode`, `FunctionId`, and serialized parameters.

The response normally uses the same accepted TCP connection. The worker does not need to open another connection to the client endpoint.

The resulting separation:

TCP layer = network framing + dispatch + concurrency gate

Facade tier = callable server operations

Domain model = enterprise/business behavior

Persistence = durable state

USER

is it possible for the listener to simply hand the unparsed incoming package to the worker thread, where the worker thread reads the first byte and acquires a `readerwriterlock` on a generic singleton object or must the listener to this, in any case, analyse and describe a full setup

ASSISTANT

Yes. The listener can be kept extremely small and hand the complete unparsed request package to a worker. In your design, I think that is cleaner than having the listener interpret the read/write byte itself.

The important distinction is between network framing and application parsing. The listener must know enough to determine where one TCP message ends, because TCP is only a byte stream. But it does not need to understand the contents of the message beyond that.

A request package could conceptually be:

`Int32 PackageLength`

`Byte AccessMode`

`Int32 FunctionId`

`Byte[] SerializedParameters`

The listener reads `PackageLength`, then keeps reading until it has exactly that many bytes. At that point it hands the whole byte array to a worker without looking at `AccessMode`, `FunctionId`, or parameters.

The worker then owns application interpretation:

- reads AccessMode and FunctionId
- deserializes parameters
- acquires ReaderWriterLockSlim
- invokes facade synchronously
- releases the lock
- serializes the result
- sends the response
- returns to the pool

The lock can simply be a single generic singleton synchronization object shared by the entire server-domain-model instance.

READ facade request -> acquire shared/read lock

WRITE facade request -> acquire exclusive/write lock

The server should still validate the requested access mode against function metadata.

If connections are persistent, the actual listener should ideally only Accept(). A connection handler owns each accepted socket, performs framing, keeps the connection alive, and hands complete packages to workers.

So:

LISTENER = establishes TCP connections

CONNECTION LAYER = owns connected socket, performs framing, keeps connection alive, hands complete packages to workers

WORKER = interprets protocol, acquires static reader/writer lock, calls facade, creates response, sends through originating

FACADE = knows nothing about sockets or locking

DOMAIN MODEL = knows nothing about TCP

PERSISTENCE = knows nothing about domain storage

USER

how does the worker thread know where to send the response to (is the client end-point an implicit element of the tcp/ip protocol?)?

i do prefer the static lock holder as i see that this way of doing it does not cross into the implementation of the facade layer. is this true? explain further with a use case on the benefit of persistent connection layer rather than the listener reading the byte array

ASSISTANT

Yes. With TCP, the worker does not need the client endpoint encoded inside the application package if the response is sent back over the same accepted TCP connection.

When the server listener accepts a client connection, the operating system gives the server a connected socket that already represents both ends of that TCP session.

The server-side socket already knows the remote IP address and remote port as part of the TCP connection state. If the listener or connection handler passes this connected socket together with the request bytes to the worker, the worker can simply send the response through that connection.

The worker should therefore receive:

Connection

+

Request

rather than only Request.

Regarding your static lock holder: yes, your interpretation is correct. A static ServerDomainModelLock can keep the locking concern entirely above the facade implementation.

The worker acquires the lock, calls the facade function, then releases the lock. The facade does not need to know that a reader/writer lock exists. This improves testability because the same facade function can be called directly in a unit test without running TCP or synchronization infrastructure.

The assumption should be explicit: one static lock protects one logical server-side domain model within that process.

Persistent connections are useful because an interactive client can connect once and perform many facade calls instead of connect/request/disconnect for every operation. This avoids repeated TCP connection establishment overhead and later avoids repeated TLS handshakes if TLS is added.

A persistent connection also provides a natural client-session communication channel.

The cleaner split is:

```
Listener:
  Accept() new connections only.

Connection handler:
  read package length
  read complete package
  submit connection + byte[] to worker
  repeat

Worker:
  parse request
  determine read/write
  deserialize arguments
  acquire static lock
  invoke facade
  release lock
  serialize result
  send through connection
```

Use case: three ENTER clients A, B, and C.

If the only listener thread accepts A and then waits while reading a large or slow request from A, B and C may wait to be accepted.

With a connection layer, the listener accepts A and hands its socket to a connection handler, immediately returns to Accept(), then accepts B and C. Each connection can receive bytes independently and submit complete requests to the shared worker pool.

The server-side TCP architecture therefore becomes:

```
LISTENER
  establishes TCP connections

CONNECTION LAYER
  owns connected socket
  performs message framing
  keeps connection alive
  hands complete packages to workers

WORKER
  interprets OPUS.NET protocol
  acquires static reader/writer lock
  calls facade
  creates response
  sends through originating connection

FACADE
  knows nothing about sockets or locking

DOMAIN MODEL
  knows nothing about TCP

PERSISTENCE
  knows nothing above domain storage
```

A low-level connection abstraction should hide partial Socket.Send() and Receive() behavior with SendAll(byte[]) and ReadExactly(n).

USER

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ASSISTANT

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